DENISH BORAD

Backend & Systems Engineer | Python (FastAPI, Django), Rust & Distributed Systems

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PROFESSIONAL SUMMARY

Performance-focused Backend & Systems Engineer with over 3 years of hands-on experience designing, building, and deploying high-concurrency backend architectures, mission-critical IoT cloud pipelines, and RESTful APIs. Deep technical expertise in Python (FastAPI, Django, Django REST Framework) and systems programming with Rust (Axum, Tokio). Proven expertise in low-latency hardware-to-cloud protocols (raw TCP sockets, WebSockets, Socket.IO, TSPL industrial printing), asynchronous background queues (Celery, Redis), relational database engineering (PostgreSQL, MySQL, complex indexing, SQLAlchemy), and multi-provider payment vaulting (Stripe, PayPal Vault, Razorpay). Experienced in manual and automated server infrastructure orchestration on Ubuntu AWS EC2 environments, utilizing Nginx reverse proxies, Gunicorn, systemd, and CI/CD automation.

TECHNICAL SKILL-SET

- **Core Languages & Frameworks:** Python, FastAPI, Django, Django REST Framework (DRF), Rust (Axum, Tokio, SQLx), Node.js, SQL, ReactJS / Vue.js (Integration)
- **Asynchronous & Distributed:** Celery, Redis (Caching & Broker), WebSockets, Socket.IO, Raw TCP/IP Socket Networking, Event-Driven Architecture
- **Databases & ORM:** PostgreSQL, MySQL, SQLite, MongoDB, SQLAlchemy, Django ORM, SQLx, Connection Pooling, Complex Query Optimization
- **IoT & Hardware Automation:** Raw TCP Sensory Telemetry Sockets, Industrial TSPL Thermal Printing (Port 9100 & Win32 FFI TSCLIB.dll), Relay Motor Control
- **Cloud, DevOps & Systems:** AWS EC2, Ubuntu/Linux Server Administration, Nginx Reverse Proxy, Gunicorn, Docker, Systemd, Git/GitHub, CI/CD, SSL/Certbot
- **Third-Party APIs & Payments:** Meta WhatsApp Cloud/Business API, Stripe, PayPal REST/Vault, Razorpay API, FCM (v1 HTTP), Google Distance Matrix API
- **AI & Intelligent Features:** YOLOv8 Computer Vision (ANPR/FastALPR microservice), LLM API Integrations (OpenAI ChatGPT, Google Gemini)

PROFESSIONAL WORK EXPERIENCE

Backend & Systems Engineer | Inlancer Technologies Pvt Ltd

Aug 2022 – Present | Rajkot, India

- **Scalable API Architecture:** Architected, developed, and deployed over 15+ production-grade REST APIs, asynchronous microservices, and IoT communication pipelines powering enterprise operations across logistics, housing ERP, dining, and aquaculture.
- **IoT & Real-Time Engineering:** Pioneered real-time socket communication architectures using raw TCP sockets, WebSockets, and Socket.IO, maintaining persistent, low-latency duplex connections with distributed IoT edge panels and hardware sensors.
- **Asynchronous Systems:** Implemented Celery distributed worker pools backed by Redis brokers for asynchronous dispatching, image rasterization, automated SMS/WhatsApp alerts, and batch reporting, achieving a 40%+ reduction in endpoint latency.
- **Fintech & Payment Vaulting:** Engineered secure multi-provider checkout and recurring billing pipelines utilizing Stripe API, PayPal Vault tokens, and Razorpay cryptographically signed webhooks, guaranteeing zero payment leakage.
- **Cloud DevOps & Linux Admin:** Managed production cloud infrastructure on Ubuntu AWS EC2 instances; configured Nginx reverse proxies with SSL/TLS termination, automated Gunicorn process management via systemd, and streamlined deployments with GitHub Actions.

Backend Developer Intern | Inlancer Technologies Pvt Ltd

Jan 2022 – Jul 2022 | Rajkot, India

- **API Development:** Engineered clean, test-driven RESTful APIs using Python, Django, and Django REST Framework for web and mobile consumer clients.
- **Database Modeling:** Designed and maintained normalized relational databases (PostgreSQL & MySQL); constructed efficient queries, constraints, and database indexing.
- **Agile Collaboration:** Collaborated with cross-functional teams using Git workflows, performing code reviews, Postman API documentation, and unit test suites.

KEY PRODUCTION PROJECTS UNDERTAKEN

Society Mitr (सोसायटी मित्र) — High-Performance Housing Society ERP & AI Gatekeeping

Tech Stack: *Rust (Axum, Tokio, SQLx), PostgreSQL, Python (FastAPI, ONNX Runtime, YOLOv8), Razorpay, Docker*

- **High-Performance Systems:** Architected high-throughput multi-tenant ERP backend in Rust with deterministic sub-millisecond API response times, zero-cost memory safety, and native connection pooling.
- **AI Gatekeeping Microservice:** Engineered dedicated Python computer-vision microservice running YOLOv8/FastALPR for real-time automated number plate recognition (ANPR) and barrier gate triggering.
- **Security & Billing:** Implemented dynamic hierarchical RBAC with per-user permission overrides, automated PostgreSQL quota triggers, and cryptographically verified Razorpay subscription billing.

ThinkX Panic Alarm System — Enterprise IoT Cloud Backend & Edge Gateway

Tech Stack: *Python (FastAPI), Rust (Tokio/Axum), PostgreSQL, Redis, Celery, WebSockets, Raw TCP Sockets*

- **Mission-Critical IoT:** Built mission-critical IoT platform coordinating physical panic panels, wireless RF repeaters, and edge sensors across educational campuses and commercial facilities.
- **Real-Time Telemetry:** Overcame restrictive NAT/firewalls using continuous two-way raw TCP & WebSocket persistent channels, delivering sub-second alert escalation and spatial floor-map tracking.

OpenUp Portal — On-Demand Emergency Roadside Assistance & Lockout Dispatch Platform

Tech Stack: *Python (Django, DRF), Celery, Redis, MySQL, Google Distance Matrix API, Firebase Cloud Messaging (FCM v1), Stripe, PayPal Vault*

- **Automated Dispatching:** Constructed on-demand dispatch engine (Uber-like model) dynamically matching stranded motorists with nearby mobile locksmiths via Google Distance Matrix API proximity calculations.
- **Fintech Vaulting:** Built multi-gateway tokenized payment vaulting (Stripe, PayPal Vault, Venmo, Apple Pay) enabling zero-touch automatic capture upon technician job sign-off.

Liliya API — Enterprise WhatsApp Marketing & Conversational Automation Platform

Tech Stack: *FastAPI (ASGI), Celery, Redis, Socket.IO, MySQL, Meta WhatsApp Cloud API, OpenAI ChatGPT API, Google Gemini API*

- **High-Throughput SaaS:** Architected asynchronous SaaS backend capable of dispatching thousands of personalized campaign messages per minute through official Meta WhatsApp Cloud API.
- **Conversational AI & Inbox:** Developed visual chatbot workflow execution engine, collaborative multi-agent live chat inbox via Socket.IO, and AI auto-response fallbacks using ChatGPT & Gemini APIs.

Industrial Thermal Label & Barcode/QR Printing Desktop Suite

Tech Stack: *Python 3, PyQt5, Ctypes (Win32 FFI & TSCLIB.dll), Linux CUPS Raw Spooling, TCP Sockets (TSPL Port 9100), CairoSVG, Pillow*

- **Hardware Automation:** Engineered industrial hardware automation suite communicating directly with TSC thermal printers via native Win32 FFI and raw network sockets, eliminating driver print-spool lag.
- **Micro-Precision Printing:** Built high-precision vector-to-1-bit PCX graphic pipeline for micro-millimeter printing on miniature 84A-12mm jewellery tags and automotive warranty labels.

Tequila's Mexican Restaurant — Multi-Location Loyalty Rewards Ledger & Operations

Tech Stack: *Python (Django & DRF), MySQL, Redis, Celery, Firebase Cloud Messaging (FCM v1), Custom Dynamic RBAC*

- **Rewards Accounting:** Engineered double-entry style points ledger regulating dining rewards in strict 100-point blocks, with dual-mode anti-fraud validation (GPS check-ins & receipt photo auditing).
- **Geo-Engagement:** Built automated geo-targeted push notification pipelines via Celery workers and FCM v1 to alert diners who favorited specific branch locations with targeted promotions.

Sagar Aqua — Industrial IoT Aquaculture Water Monitoring Platform

Tech Stack: *FastAPI, Python-Socket.IO, Raw TCP/IP, MySQL, SQLAlchemy, ReactJS*

- **Industrial Telemetry:** Engineered continuous environmental telemetry aggregation backend collecting dissolved oxygen and temperature metrics from pond-side hardware sensors via raw TCP.
- **Automated Motor Control:** Designed automated starter/soft-starter aerator motor trigger sequences based on real-time sensory thresholds with fail-safe hardware overrides.

EDUCATION

Master of Science in Information Technology (M.Sc. IT) **2024 – 2026**

Saurashtra University (Sarvodaya College) | Rajkot, India | **CGPA: 8.71 / 10.0**

Bachelor of Computer Applications (BCA) **2019 – 2022**

Saurashtra University (H & H B Kotak Institute) | Rajkot, India | **CGPA: 7.90 / 10.0**

PROFESSIONAL CERTIFICATIONS

- Python for Enterprise Web Development — Inlancer Technologies LLP
- Django REST Framework (DRF) Production Architecture — Inlancer Technologies LLP
- FastAPI: High-Performance Asynchronous Python Frameworks — Inlancer Technologies LLP